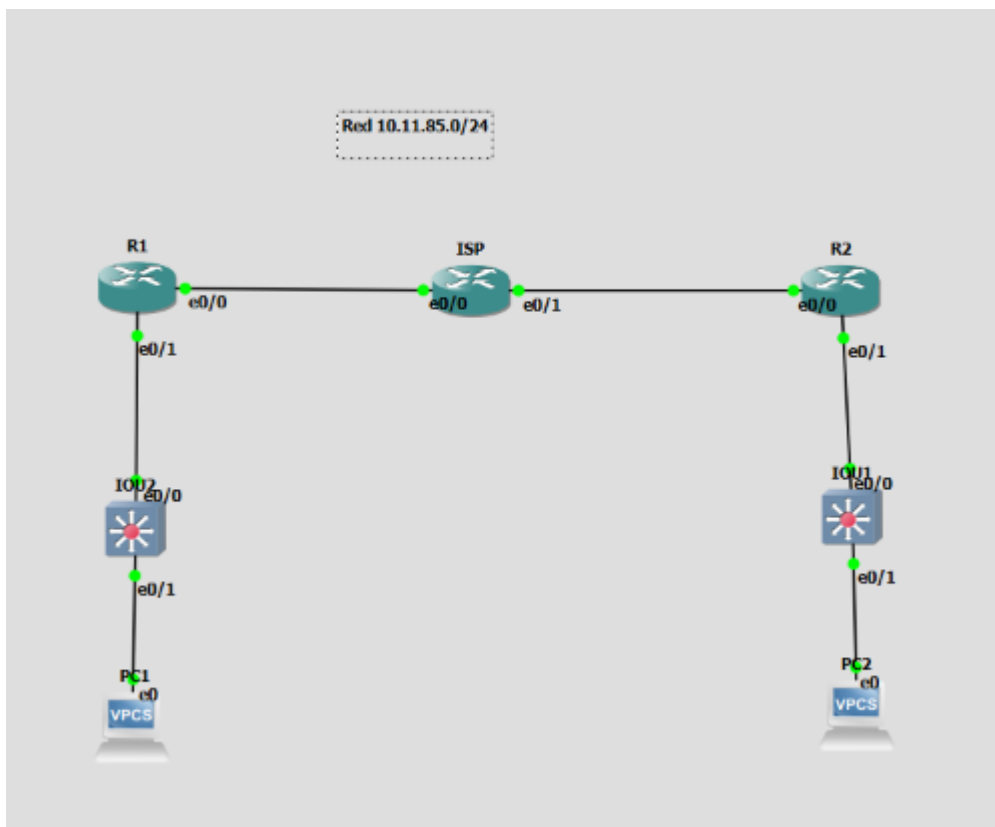


1. Objetivo

Implementar VPN Site-to-Site con **IPSec IKEv2 route-based (VTI)**. La diferencia respecto a IKEv1 route-based es que el **crypto ipsec profile** ahora incluye **set ikev2-profile PROF-IKEV2**, vinculando la autenticación IKEv2 directamente al túnel virtual. No se usa crypto map ni ACL.

2. Topología



Parámetro	Valor
Proposal	PROP-IKEV2
Cifrado	AES-CBC-256
Integridad	SHA-256
Grupo DH	(2048-bit)

Keyring / PSK	KR-IKEV2 / ITLA2024
Profile	PROF-IKEV2

Direccionamiento IP

Dispositivo	Interfaz	Dirección IP	Descripción
R1	e0/0	10.11.85.65/26	WAN → ISP
R1	e0/1	10.11.85.1/26	LAN → SW1
R1	Tunnel0	10.10.10.1/30	VTI hacia R2
ISP	e0/0	10.11.85.66/26	Hacia R1
ISP	e0/1	10.11.85.129/26	Hacia R2
R2	e0/0	10.11.85.130/26	WAN → ISP
R2	e0/1	10.11.85.193/26	LAN → SW2
R2	Tunnel0	10.10.10.2/30	VTI hacia R1
PC1	e0	10.11.85.2/26	GW: 10.11.85.1
PC2	e0	10.11.85.194/26	GW: 10.11.85.193

3. Parámetros

Auth local/remota	Pre-Shared Key
Parámetro	Valor

Transform Set	TS-IKEV2-VTI / ESP / Tunnel / AES-256 / SHA256
IPSec Profile	PROF-VTI-IKEV2
set ikev2-profile	PROF-IKEV2 (dentro del ipsec profile)
Tunnel mode	ipsec ipv4
Tunnel protection	ipsec profile PROF-VTI-IKEV2

4. Configuración R1

R1

enable
configure terminal

hostname R1

interface Ethernet0/0
description WAN hacia ISP
ip address 10.11.85.65 255.255.255.192
no shutdown

interface Ethernet0/1
description LAN hacia SW1-PC1
ip address 10.11.85.1 255.255.255.192
no shutdown

ip route 0.0.0.0 0.0.0.0 10.11.85.66

crypto ikev2 proposal PROP-IKEV2
encryption aes-cbc-256
integrity sha256
group 14

crypto ikev2 policy POL-IKEV2
proposal PROP-IKEV2

crypto ikev2 keyring KR-IKEV2
peer R2
address 10.11.85.130
pre-shared-key ITLA2026
!

```
crypto ikev2 profile PROF-IKEV2
match identity remote address 10.11.85.130 255.255.255.255
authentication remote pre-share
authentication local pre-share
keyring local KR-IKEV2

crypto ipsec transform-set TS-IKEV2-VTI esp-aes 256 esp-sha256-hmac
mode tunnel

crypto ipsec profile PROF-VTI-IKEV2
set transform-set TS-IKEV2-VTI
set ikev2-profile PROF-IKEV2

interface Tunnel0
description VTI hacia R2 (IKEv2)
ip address 10.10.10.1 255.255.255.252
tunnel source Ethernet0/0
tunnel destination 10.11.85.130
tunnel mode ipsec ipv4
tunnel protection ipsec profile PROF-VTI-IKEV2
no shutdown

ip route 10.11.85.192 255.255.255.192 Tunnel0

end
```

5. Configuración R2

```
enable
configure terminal
hostname R2
interface Ethernet0/0
description WAN hacia ISP
ip address 10.11.85.130 255.255.255.192
no shutdown

interface Ethernet0/1
description LAN hacia SW2-PC2
ip address 10.11.85.193 255.255.255.192
no shutdown
```

```

ip route 0.0.0.0 0.0.0.0 10.11.85.129
crypto ikev2 proposal PROP-IKEV2
  encryption aes-cbc-256
  integrity sha256
  group 14
crypto ikev2 policy POL-IKEV2
  proposal PROP-IKEV2
crypto ikev2 keyring KR-IKEV2
  peer R1
    address 10.11.85.65
    pre-shared-key ITLA2026
  !
crypto ikev2 profile PROF-IKEV2
  match identity remote address 10.11.85.65 255.255.255.255
  authentication remote pre-share
  authentication local pre-share
  keyring local KR-IKEV2

crypto ipsec transform-set TS-IKEV2-VTI esp-aes 256 esp-sha256-hmac
mode tunnel
crypto ipsec profile PROF-VTI-IKEV2
  set transform-set TS-IKEV2-VTI
  set ikev2-profile PROF-IKEV2
interface Tunnel0
  description VTI hacia R1 (IKEv2)
  ip address 10.10.10.2 255.255.255.252
  tunnel source Ethernet0/0
  tunnel destination 10.11.85.65
  tunnel mode ipsec ipv4
  tunnel protection ipsec profile PROF-VTI-IKEV2
  no shutdown
ip route 10.11.85.0 255.255.255.192 Tunnel0
end

```

6. Configuración ISP

```

enable
configure terminal
hostname ISP

interface Ethernet0/0
  ip address 10.11.85.66 255.255.255.192
  no shutdown

```

```
interface Ethernet0/1
ip address 10.11.85.129 255.255.255.192
no shutdown

end
write memory
```

7. Verificación

```
! Estado SA IKEv2 (QM_IDLE → READY en IKEv2)
show crypto ikev2 sa

R1#sh crypto ikev2 sa
IPv4 Crypto IKEv2 SA

Tunnel-id Local Remote fvrf/ivrf Status
1 10.11.85.65/500 10.11.85.130/500 none/none READY
Encr: AES-CBC, keysize: 256, Hash: SHA256, DH Grp:14, Auth sign: PSK, Auth verify: PSK
Life/Active Time: 86400/973 sec

IPv6 Crypto IKEv2 SA
```

```
show crypto ikev2 sa detailed
```

```
R1#show crypto ikev2 sa detailed
IPv4 Crypto IKEv2 SA

Tunnel-id Local Remote fvrf/ivrf Status
1 10.11.85.65/500 10.11.85.130/500 none/none READY
Encr: AES-CBC, keysize: 256, Hash: SHA256, DH Grp:14, Auth sign: PSK, Auth verify: PSK
Life/Active Time: 86400/1001 sec
CE id: 1002, Session-id: 1
Status Description: Negotiation done
Local spi: AC64C866324DBC55 Remote spi: E3574A3F881668A3
Local id: 10.11.85.65
Remote id: 10.11.85.130
Local req msg id: 2 Remote req msg id: 0
Local next msg id: 2 Remote next msg id: 0
Local req queued: 2 Remote req queued: 0
Local window: 5 Remote window: 5
DPD configured for 0 seconds, retry 0
Fragmentation not configured.
Extended Authentication not configured.
NAT-T is not detected
Cisco Trust Security SGT is disabled
Initiator of SA : Yes

IPv6 Crypto IKEv2 SA
```

```

R1#show crypto ikev2 profile
IKEv2 profile: PROF-IKEV2
Ref Count: 5
Match criteria:
  Fvrf: global
  Local address/interface: none
Identities:
  address 10.11.85.130 255.255.255.255
Certificate maps: none
Local identity: none
Remote identity: none
Local authentication method: pre-share
Remote authentication method(s): pre-share
EAP options: none
Keyring: KR-IKEV2
Trustpoint(s): none
Lifetime: 86400 seconds
DPD: disabled
NAT-keepalive: disabled
Ivrf: none
Virtual-template: none
AAA EAP authentication mlist: none
AAA Accounting: none
AAA group authorization: none
AAA user authorization: none
R1#

```

```

R1#show crypto session
Crypto session current status

Interface: Tunnel0
Profile: PROF-IKEV2
Session status: UP-ACTIVE
Peer: 10.11.85.130 port 500
[Session ID: 1
IKEv2 SA: local 10.11.85.65/500 remote 10.11.85.130/500 Active
IPSEC FLOW: permit ip 0.0.0.0/0.0.0.0 0.0.0.0/0.0.0.0
Active SAs: 2, origin: crypto map

```

show crypto ikev2 proposal

```

R1#show crypto ikev2 proposal
IKEv2 proposal: PROF-IKEV2
Encryption : AES-CBC-256
Integrity : SHA256
PRF : SHA256
DH Group : DH_GROUP_2048_MODP/Group 14
IKEv2 proposal: default
Encryption : AES-CBC-256 AES-CBC-192 AES-CBC-128
Integrity : SHA512 SHA384 SHA256 SHA96 MD596
PRF : SHA512 SHA384 SHA256 SHA1 MD5
DH Group : DH_GROUP_1536_MODP/Group 5 DH_GROUP_1024_MODP/Group 2
R1#

```

! SA IPsec + contadores encrypt/decrypt

show crypto ipsec sa

```

AAA user authorization: none
R1# show crypto ipsec sa | include pkts
#pkts encaps: 15, #pkts encrypt: 15, #pkts digest: 15
#pkts decaps: 15, #pkts decrypt: 15, #pkts verify: 15
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
R1#

```

```

show interface
tunnel0

```

```
R1#sh ip int bri | include Tunnel0
Tunnel0      |10.10.10.1      YES manual up
R1#
```

show ip route

```
R2#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override

Gateway of last resort is 10.11.85.129 to network 0.0.0.0

S*    0.0.0.0/0 [1/0] via 10.11.85.129
      10.0.0.0/8 is variably subnetted, 7 subnets, 3 masks
C      10.10.10.0/30 is directly connected, Tunnel0
L      10.10.10.2/32 is directly connected, Tunnel0
S      10.11.85.0/26 is directly connected, Tunnel0
C      10.11.85.128/26 is directly connected, Ethernet0/0
L      10.11.85.130/32 is directly connected, Ethernet0/0
C      10.11.85.192/26 is directly connected, Ethernet0/1
L      10.11.85.193/32 is directly connected, Ethernet0/1
R2#
```

Comandos trace y
ping, verificación
del salto

```
PC1> ping 10.11.85.194

64 bytes from 10.11.85.194 icmp_seq=1 ttl=62 time=2.324 ms
64 bytes from 10.11.85.194 icmp_seq=2 ttl=62 time=3.112 ms
64 bytes from 10.11.85.194 icmp_seq=3 ttl=62 time=1.433 ms
64 bytes from 10.11.85.194 icmp_seq=4 ttl=62 time=1.793 ms
64 bytes from 10.11.85.194 icmp_seq=5 ttl=62 time=2.238 ms

PC1> trace 10.11.85.194
Trace to 10.11.85.194, 8 hops max, press Ctrl+C to stop
 1  10.11.85.1    0.397 ms  0.428 ms  0.504 ms
 2  10.10.10.2    1.432 ms  0.995 ms  0.925 ms
 3  *10.11.85.194  3.202 ms (ICMP type:3, code:3, Destination port unreachable)
```